Chapter 1: Brand Foundation

1. Brief About Tree'd and Core Purpose

Tree’d History Guide is building a new category of museum technology: conversational audio guides that visitors can interact with naturally, without screens, apps, or technical onboarding. Using a dedicated handset powered by Conversational AI, Tree’d enables visitors to explore at their own pace, tap to hear curated stories, and ask follow-up questions in their preferred language.

Core Purpose

Tree’d exists to make high-quality cultural interpretation accessible, personal, and multilingual. Traditional tools force museums to choose between three imperfect options:

1. Static audio guides that deliver one-way scripts and cannot adapt to visitor curiosity.
2. Live guided tours that are immersive but costly, difficult to schedule, never self-paced, and not available in multiple languages.
3. Visually immersive but ultimately counterproductive in a gallery setting. They pull the visitor’s attention away from the exhibition itself, introduce a technological barrier (downloads, sign-ins, device compatibility), and often require setup or staff intervention.

Tree’d closes this gap. Its system introduces affordable, two-way conversation that respects the rhythm of each visitor. The experience is self-paced, intuitive, and available across major languages, giving visitors the freedom to explore and the ability to receive instant, accurate answers.

Technology and Experience

Tree’d’s Conversational AI is trained exclusively on museum-validated knowledge bases. It recognizes context, recalls exhibit information, and responds in real time without exposing visitors to screens or external distractions. The result is a guided experience that feels personal, intuitive, and culturally respectful.

Naming Convention

While the company is known as Tree'd, the full name is **Tree'd History Guide**. The inclusion of "History Guide" in the logo is an **intentional measure** to **avoid any misinterpretations of the name**. For example, this clarification prevents confusing the business with net-zero initiatives, plant nursery, or farming operations. The logo architecture itself is based on the shape of **Tree'd's Handset**.

Founding

The company was established in **2022** in Amsterdam and required **years of R&D** to reach its current stage.

2. Vision, Mission, and Commitment

Vision

The overarching vision is **to make culture & history exploration feel personal, everywhere**. Tree'd imagines a world where every visitor can **explore as deep as they want, ask as many questions as their curiosity takes them**, and **connect more with cultures they see**. This vision exists to make that connection possible **across languages, borders, and generations**. The premise is that culture is not static, and therefore museum experiences should not be static either.

Mission

The mission of Tree'd is **to build, develop, and implement intuitive, multilingual solutions** that enable museums to offer **more immersive, accessible, and engaging experiences**.

Commitment

Tree'd's commitment stems from the belief that **Museums Deserve More Than Static Playback**. Therefore, the commitment is **to build tech solutions that respect culture & history**.

3. Foundational Values

Tree'd is guided by four fundamental values in its design and operation:

1. **Respect Over Disruption:** Tree’d designs with **immense respect for cultural spaces and the stories they hold**. The technology's purpose is to **nurture the visitor-exhibit relationship** and **never interrupt it**.

2. **Curiosity Comes First:** The core belief is that **questions are where the experience begins**. Tree’d is explicitly **built to reward curiosity**. This applies to both the visitor in a gallery and the designer on the internal team.

3. **Simple Yet Powerful:** Tree’d believes that in the cultural space, **the best ideas don’t need to announce themselves**. The company builds intuitive, powerful products that are **understood in seconds, not explained in steps**.

4. **Build With Intent:** Because Tree’d works in spaces where **silence matters and distraction isn’t an option**, the company builds with care, ensuring that **every detail must serve the overall experience**.

4. Contact Details

- **Primary Office Address (Netherlands):** Plantage Kerklaan 21, 1018 SZ Amsterdam, Netherlands.
- **Secondary Office (Egypt):** Cairo, Egypt (based on previous conversation history).
- **Phone Number:** +31 6 87600185.
- **Email:** hello@treed.co.

Chapter 2: The Market Problem: Why Existing Museum Technologies Fall Short

Tree'd was built to address a "massive gap in the market" that exists between the current passive museum experience and the ideal immersive, conversational experience where visitor curiosity is actually rewarded, not just shut down. The truth is that, currently, the museum experience is "fundamentally broken for millions and millions of visitors".

The core issue is that visitors are forced to choose between three undesirable outcomes: the experience is either **one-way, inflexible, or overcomplicated**. This failure boils down to three main types of technological shortcomings that are "flat out failing" museums.

1. Problem Statement: The Massive Gap in the Market

The fundamental problem is a disparity between what is currently offered and what visitors crave.

- **Current State:** Today's technology primarily consists of **passive audio guides** and **super distracting apps** that pull visitors away from the art. These solutions are non-immersive and one-sided.
- **Ideal State:** The desire is an **immersive conversational experience** where a visitor's **curiosity is actually rewarded**.
- **The Gap:** This "massive gap" represents a passive opportunity that Tree'd is seizing, as it is the space where two-way, personal, and real-time conversations can be introduced. Tree'd sets out to provide the freedom to explore at their desired pace, the ability to ask questions, and the satisfaction of getting an answer—instantly and in the visitor's language.

2. Why Classic Passive Audio Guides Are Not Doing a Good Enough Job

Traditional or classic audio guides, while offering some benefits, fail at the crucial point of visitor engagement and curiosity.

1. **Failure to Fulfill Curiosity:** The most significant flaw is that they **can't answer a question**. Visitors **cannot ask a question** or follow a **sudden spark of curiosity**. This inability to fulfill curiosity means they fail to feel personal and give knowledge a visitor is seeking.
2. **Static and Generic Content:** Traditional audio guides simply play **pre-recorded scripts**. This creates a **one-way street** where the content is passive and generic. They deliver pre-recorded lectures, which limits customizing the depth of cultural exploration.

3. Why Live Tour Guides Are Not Suitable for All

Live human guides offer certain immersive qualities, but they introduce major barriers related to cost, rigidity, and independence.

1. **Cost:** Live guides **offer great depth, sure, but they're really expensive**. This cost often makes the experience inaccessible to all visitors.

2. **Rigidity and Lack of Self-Pace:** Live guides are "super rigid". They are **never self-paced**. Visitors are **stuck in a group, following someone else's rhythm**, which **totally destroys that personal, self-paced journey** that visitors crave.
3. **Required Scheduling:** Unlike Tree'd, which requires **no scheduling**, live tours mandate adherence to a timeline (i.e., specific tour start time, specific tour duration)

The market forced visitors to choose between an affordable, self-paced, but passive and generic experience (traditional audio guides) or an immersive experience that was costly, rigid, and required following someone else's pace (live guides). Tree'd sought to combine the best of both: **Affordable, self-paced, immersive, and readily available tours**.

4. Why AR/VR and Similar Technologies Are Double-Edged Swords (The Immersive Tech Failure)

AR, VR, and mobile apps aim to enhance museum interpretation, but they conflict with two core principles of how visitors actually experience culture inside a gallery.

1. They Compete With the Visitor's Most Important Sense: Sight

A museum is built around a primary sensory experience: sight.

The entire architecture of a gallery is designed to direct the visitor's eyes toward the collection. Sight is meant to be the primary sense, while other senses gently support it. Sound can guide. Context can enrich. But nothing should compete with what the eyes are there to absorb.

Screen-based technologies invert that hierarchy.

- They introduce new visual layers that compete with the work itself.
- They pull the visitor's gaze away from the object and toward interfaces, effects, or overlays.
- They add stimulus to the very sense that museums carefully protect and curate.

Instead of deepening the sensory experience, these tools compete with it, turning the visitor's attention away from the object they came to see.

2. They Create a Technological Barrier for Many Visitors

AR/VR systems and mobile apps assume a level of digital fluency that not all visitors possess.

- Some interfaces can add friction.
- Non-tech-familiar individuals often struggle with the setup.
- Rather than making interpretation more inclusive, these tools risk making it less accessible.

Tree’d removes these drawbacks by taking the best qualities of the three existing approaches—audio guides, live tours, and digital immersion—while stripping away everything that makes them problematic. Like traditional audio guides, Tree’d is affordable, self-paced, and easy to use, but it replaces their passivity with real conversational intelligence. It offers the depth and richness of a live guide without the cost, scheduling rigidity, or dependence on group pacing. And it delivers the contextual richness people seek from modern digital tools, yet without screens, visual clutter, or technological friction. By grounding the experience in sound rather than sight, Tree’d complements the visitor’s primary sensory engagement instead of competing with it. The result is a single, coherent solution that preserves all the advantages of current systems while eliminating the barriers that keep them from delivering a truly personal, accessible, and focused museum experience.

Chapter 3: The Tree’d Solution: Architecture and Operation

Tree’d is a fully integrated museum technology ecosystem that redefines how visitors interact with culture. It is designed as a **self-contained, AI-powered audio guide system** that combines intuitive hardware, conversational AI, and data-driven insight.

1. Brief About Our Solution (Core Concept)

Tree’d introduces a **whole new category**: the next-generation conversational audio experience. The core promise is to fuse the affordability of a traditional audio guide with the responsive, intelligent depth of a live expert.

- **The Core Product and Functionality:** Tree’d is an **AI-powered audio guide** that enables museum visitors to explore at their own pace, ask questions, and get instant answers—**without screens or apps**. It transforms the traditional audio guide model into an **interactive, self-paced system** that encourages curiosity rather than prescribing a fixed route.
- **Two-Way Conversation:** Tree’d brings two-way conversation to spaces that have always been served by one-sided, non-immersive solutions. Unlike static pre-recorded devices or human-guided tours, Tree’d uses **Conversational AI to create dynamic, two-way interactions** between visitors and exhibits.
- **Core Promise:** The solution creates an **immersive two-way conversation with art**. It does this while **getting rid of any and all visual distractions**.
- **Accessibility and Scalability:** The solution supports **over a dozen languages** and **scales effortlessly**, making deeper engagement possible for every visitor, in every collection. Tree’d developed its Conversational AI model to speak any major language and learn about any artifact.
- **Operational Simplicity:** The system is a **plug-and-play system** with **no screens, no apps, and no extra staff** required for operation. It delivers an experience that feels natural, immediate, and deeply informative while requiring **almost zero operational effort from museums**.

2. How It Works (The 3-Step Customer Journey)

Tree'd is designed to feel familiar from the moment a visitor picks it up, requiring **no downloads, no setup, and no onboarding**. The visitor journey is a simple, three-step process, referred to as **START. TAP. TALK.**. This system is "truly plug-and-play curiosity".

The journey begins once a visitor collects the Handset from the museum's point of sale, with **no setup required**.

Step 1: Select Your Language

- Visitors start by tapping their Handset against an **NFC Language Panel** to choose their preferred language for the journey.
- Tree'd currently supports **over 12 languages (English, Dutch, Arabic, French, Italian, German, Spanish, Portuguese, Japanese, Mandarin, Korean, Ukrainian, and Russian)**.

Step 2: Tap an Exhibit (Choose an Exhibit)

As visitors explore, they tap an **NFC tag** located near an exhibit.

- This NFC-triggered interface instantly triggers the playback of a **curated audio story** behind the piece. This audio comes from museum-approved sources. The audio itself goes through multiple rounds of approvals with the museum.

Step 3: Start a Conversation

The visitor journey transforms into a conversation: after listening to the default audio track, visitors can **ask follow-up questions verbally**.

- The Handset listens, understands what the visitor is asking, and responds. The **AI recognizes context**, and processes questions through Tree'd's **Conversational AI layer** to generate responses in **real-time**. What starts as a tour becomes a conversation.

3. The Building Blocks of Our System

Tree'd is a **complete, scalable ecosystem**. The system is built on intuitive hardware, Conversational AI, and reliable NFC to create a seamless visitor experience.

Collectively, the hardware components—the Handsets and the Tree—are called the **"Audio Guide System"**.

A. The Handset (Physical Interface)

The Handset is the device visitors use to interact with the museum's content. It is designed to be immediately understandable, distraction-free, and reliable for continuous use throughout a full museum day.

1. Design and Form Factor

- Shape: Modeled after a vintage telephone handset to provide instant familiarity and ergonomic comfort.
- Interface: Screen-free and buttonless. Visitors only lift it, listen, speak, and tap NFC markers.
- Simplicity: The absence of screens and buttons is intentional to eliminate distractions and ensure a zero-learning-curve experience for visitors across all age groups.
- Wrist Loop: The handset includes a coiled telephone-style wrist loop at its base, allowing visitors to comfortably hang the device from their wrist as they move through the museum. This ensures ease of movement, reduces the risk of accidental drops, and keeps the handset accessible throughout the visit.

2. Materials and Build

- Body Material: 3D-printed using PLA (Polylactic Acid), a biodegradable, plant-based polymer.
- Finishing: Each unit is handcrafted: sanded, primed, and painted after printing to achieve a smooth, durable, museum-grade finish.
- Safety: All paints and coatings are non-toxic and compliant with common safety standards for handheld public devices.
- Durability: The structure is designed to withstand high-frequency daily use, accidental drops, and repeated sanitation.

3. Core Functionality

- Conversational AI: The Handset answers visitor questions in real time.
- Language Support: It can understand and speak more than a dozen languages.

- NFC-Activated: Visitors tap the handset onto an NFC tag placed next to an artwork or exhibit to start the experience. The handset has an NFC reader that reads up to 5 cm.

4. Audio and Feedback System

- Speaker: Tuned for clarity in galleries, reducing sound spill into the surrounding space.
- Haptic Feedback: A subtle vibration confirms NFC scans or successful recognition of a spoken question.
- Micro-LED Indicators: Monitor battery status, network connectivity, and power.
- Microphone: A digital microphone captures the visitor's speech as sound waves, converts them into digital audio signals, and sends them to the speech-to-text engine, which transforms the audio into text that the language model can interpret.

5. Performance and Power

- Battery: Lithium-ion cell supporting up to 6 hours of continuous, active use.
- Recharging: Recharges to approximately 50 percent less than an hour when placed in the Tree (the charging hub). So, <1 hour of charging = 3 hours of continuous active use.
- Thermal Management: Internal components are configured to maintain stable performance during continuous high-traffic usage.

6. Connectivity

- Wireless Protocol: Low-energy Wi-Fi connection to synchronize anonymous usage data with the Dashboard.
- Security: No personal visitor information or biometrics are stored on the device.
- Network: Connected via a local network to the hub (The Tree), where it acquires information from and sends data to.

7. Variants

- Standard Edition: Speaker-first design suitable for most gallery environments.
- Aux Edition: Includes a 3.5 mm headphone jack for institutions requiring a headphone-compatible or silent-environment edition. This version is a simple box-shaped design to be slipped into a visitor's pocket if they don't want to keep holding an audio guide for the entirety of the tour.

8. Design Intent Summary

The Handset is intentionally: Familiar, Screen-free, Distraction-free, Accessible, Durable, Museum-friendly, Reliable for full-day operation.

Its purpose is to let visitors focus their sight entirely on the cultural experience while enabling museums to deliver a modern, conversational, multilingual audio experience without screens or complexity that nurtures and complements the sight experience through hearing.

B. The Tree (Charging and Display Hub)

The Tree is the charging, storage, and synchronization hub for the Handsets. It is designed to support daily museum operations by keeping devices charged, organized, and automatically connected to the Dashboard.

1. Core Purpose

- Provide handsets with access to the knowledge database.
- Charge multiple Handsets simultaneously.
- Store and beautifully display Handsets in an organized, visitor-ready format.
- Sync usage and health data from each device to the Dashboard.
- Provide a clean, stable, low-maintenance operational base for the system.

2. Design and Form Factor

- Sculptural Structure: The Tree is a freestanding hardware unit designed to present Handsets neatly and visibly.
- Material Options: The external shell can be manufactured in wood, matte metal, or composite finishes to match museum aesthetics. Each design is tailor-made for each museum to match the theme of the museum.
- Modularity: Units are modular and can be configured as single columns, dual columns, or multi-column clusters depending on required capacity and available space.
- Space Efficiency: Designed for high-density storage while maintaining a minimal footprint.

3. Charging System

- Multi-Port Charging Matrix: Supports simultaneous charging for multiple Handsets at once.
- Rapid Recharging: The system delivers enough power for Handsets to recover approximately 50 percent charge in under an hour.
- Automatic Power Management: Each slot regulates power intake independently to protect battery health and ensure consistent performance.
- Thermal Management: Internal components are engineered to maintain stable temperatures under continuous operation.

4. Data Synchronization

- Automatic Syncing: When Handsets are docked, usage data, battery health, and error logs are automatically transferred to the Dashboard.
- Low-Energy Connectivity: Uses a local network over Wi-Fi to securely send aggregated, non-personal operational data.
- Offline Resilience: If network connectivity is unavailable, data is queued and synced once the connection is restored.

5. Hygiene and Maintenance

- Optional UVC Chamber: A UVC sterilization module can be included for museums that require sanitation between uses.
- Easy Cleaning: Exterior surfaces are smooth and durable, designed for fast wipe-down cleaning.
- Serviceability: Internal components are accessible for maintenance and repairs, which will be fully handled by Tree'd if needed.

6. Installation and Requirements

- Plug-and-Play: Installation typically requires only a standard electrical outlet.
- No Complex Infrastructure: No permanent fixtures, floor work, or networking installations are typically necessary.
- Expandable: For larger museums, additional Trees can be added and chained together for higher capacity.
- Small-Site Version: Compact, low-capacity designs are available for small galleries or travelling exhibitions.

7. Operational Role

The Tree functions as the operational backbone of the system (hence our company name) by:

- Ensuring all Handsets begin each day fully charged.
- Maintaining an organized presentation for visitor pickup.
- Automatically syncing performance and usage metrics.

C. The Dashboard (Analytics and Management Portal)

The Dashboard is a browser-based analytics and management workspace that gives museum teams visibility into how visitors interact with content and how the Tree'd system performs operationally. It consolidates engagement insights, device health, and language trends into one simple to use, yet powerful interface (just like everything Tree'd does).

1. Core Purpose

I. **Centralize all engagement and operational data into a single, unified interface:** Curators, administrators, and technical staff can monitor visitor interactions, AI question patterns, content completion rates, language usage, handset status, and overall system performance without switching between tools or relying on manual logs.

II. **Enable data-led decisions across content, exhibitions, and operations:** The Dashboard doesn't just display metrics — it reveals patterns that help museums refine how they curate and manage the visitor experience. For example:

- **Exhibition performance:** Artifact-level scan rates and completion data reveal which pieces draw attention and which may need stronger storytelling or better placement.
- **Audience segmentation:** Language distribution per artifact shows how different visitor groups engage with different themes, helping refine interpretive priorities.
- **Accessibility planning:** Insights into which languages visitors actually use guide decisions about multilingual signage, multilingual staff placement, and language offerings.
- **Future investments with Tree'd:** Demand for certain languages, high curiosity around particular collections, or repeated follow-up questions help museums decide where to expand content, deepen narratives, or introduce new exhibitions supported by Tree'd.

By turning raw interactions into structured insights, the Dashboard helps museums continuously improve storytelling quality, visitor satisfaction, and operational efficiency.

2. Data Management and Tools

2.1. Filtering and Reporting

- Filter metrics by date range, artifact, language, or exhibit.
- Export reports in CSV or PDF formats.
- Create custom dashboards for monitoring weekly, monthly, or exhibition-specific performance trends.

2.2. AI-Powered Assistance

- Summarizes patterns in visitor behavior.
- Highlights emerging interests and unusual spikes in engagement.
- Suggests potential content improvements based on AI question clusters and audio drop-offs.
- Chatbot to give key insights about performance.

2.3. Access Control and Compliance

- Hosted on a secure cloud environment, with an option to operate on local servers if data sovereignty is required.
- Supports role-based permissions (e.g., Admin, User).
- Designed to comply with GDPR; does not store personal visitor identities or biometrics.

2.4. Dashboard Metrics in Detail

I. Interactions (NFC Scans)
   - What it measures: How many times visitors tap the Handset on an NFC plate. Tracked both per session and per artifact.
     - Why it matters: Indicates the popularity and visibility of each artifact. Helps identify “high-attention drawers” vs. underperforming exhibits. Reveals whether NFC plates or signage placement is working. Useful for evaluating gallery layout, identifying blind spots, and improving wayfinding.

II. Completion Rate
   - What it measures: The percentage of audio tracks that visitors listen to at least 90% of the way through. Shown at museum-level and artifact-level.

        - Why it matters: Completion rate reflects storytelling quality, pacing, and clarity. A low completion rate on a specific artifact signals: The story is too long; The script isn't engaging; The pacing or tone needs refinement. At a museum-wide level, it is a KPI for the overall effectiveness of the audio guide experience.

III. Language Mix
    - What it measures: The distribution of sessions by language. Tracked on museum-level and per artifact.
        - Why it matters: Helps museums understand their visitor demographics more precisely. Supports operational planning such as: Multilingual signage, Hiring or positioning multilingual staff, Tailoring seasonal exhibitions for specific visitor segments. Artifact-level language mix shows which communities engage with which content, guiding decisions related to staffing, signage, and languages to add to Tree'd.

IV. AI Conversations (Follow-Up Questions)
    - What it measures: The number and type of follow-up questions visitors ask after listening to a story. Tracked per session and per artifact.
        - Why it matters: Indicates depth of curiosity and which topics trigger further exploration. Shows which artifacts stimulate intellectual engagement. Helps curators identify where to: Add more detail to scripts, Create new labels or mini-exhibits, Prepare targeted educational materials. Patterns in AI questions often reveal emerging themes or interests among visitors.

V. Visitor Engagement Rate
    - Measures: Average number of artifacts visited per session.
        - Importance: Indicates whether visitors are exploring broadly or narrowly, supporting exhibition flow analysis.

VI. Average Session Length
    - Measures: How long visitors spend using the guide.
        - Importance: Useful for understanding dwell time, congestion planning, and designing optimal gallery routes.

VII. Artifact Popularity Rankings
    - Measures: Artifacts sorted by number of scans.

        - Importance: Guides staffing, programming, and decisions around highlight tours and exhibition flow.

VIII. Battery Status
    - Measures: Battery percentages
        - Importance: Ensures operational reliability, reduces staff workload, and prevents mid-day device downtime.

IX. Maintenance Indicators
    - Measures: Uptime, error reports, and performance anomalies.
        - Importance: Helps technicians plan preventive maintenance, reducing operational disruptions.

3. Design and Usability

- Clean, minimal interface built for quick scanning and fast access to key metrics.
- Mirrors the simplicity of Tree'd's physical devices: no clutter, no unnecessary UI distractions.
- Optimized for desktop use by museum staff during daily operations.

4. Operational Role

The Dashboard enables museums to:

- Understand how visitors engage with exhibits.
- Identify which stories perform strongly and which need revision.
- Detect shifts in visitor language needs.
- Review the health and performance of the Handset fleet.
- Use data to refine exhibitions, content pacing, and visitor flow.

The Dashboard in a Nutshell: Key Data and Metrics Tracked

The dashboard collects real-time insights from all active handsets and AI sessions:

- **Interactions: NFC scans recorded** at the **Session Level** (revealing ease-of-use and NFC accessibility) and the **Artifact Level** (revealing attention drawers and flagging under-performers).
- **Completion Rate:** Measures the **% of started tracks that users finish (90% of duration)** at the **Museum Level** (measuring overall storytelling quality) and **Per Artifact** (revealing drop-offs due to length, tone, or pacing).

- **Language Mix:** Tracks **sessions in a specific language** at the **Museum Level** (guiding staffing and signage) and **NFC scans in a specific language** at the **Artifact Level** (measuring the interest of visitor groups in certain artifacts).
- **AI Convos (Conversations):** Tracks the **follow-up questions asked through the AI layer** at the **Session Level** (quantifying curiosity and engagement depth) and the **Artifact Level** (highlighting curiosity triggers, informing future scripts and exhibits).
- **Operational Data:** Includes visitor engagement rates, average session length, **most popular artifacts** (based on NFC scans), **language distribution**, **question frequency and topic clustering**, and **device diagnostics** (battery, uptime, maintenance schedule).
- Data Management and Insights: **Filtering and Reporting:** Users can **filter by date range, language, or exhibit** and **export reports** in CSV or PDF formats. Custom visual dashboards can be designed to monitor performance across various time periods, with custom analytics ranges (daily, weekly, monthly, per exhibition).
- **AI-Powered Assistance:** An **optional AI assistant** helps summarize visitor behavior, detect emerging interests, and **recommend content updates**.
- **Access and Compliance:** The dashboard is **cloud-hosted** but can operate on **local servers for data sovereignty compliance (EU GDPR-ready)**. Access is managed through **secure multi-role accounts (Admin, Curator, Technician)**.

The dashboard's design language mirrors Tree'd's physical devices—simple yet powerful, blending aesthetic minimalism with analytical precision. The museum receives **monthly analytics and periodic content reviews** to help refine stories and visitor experiences based on real engagement data.

Chapter 4: Visitor Benefits and Experience with Tree’d

The overall visitor experience is one of **"plug-and-play curiosity"**. There is **no app to download, no account to create, and no setup is required** once the visitor collects the Handset.

1. Dynamic and Conversational Engagement

The fundamental difference Tree'd offers is the transition from a monologue to a two-way dialogue.

- **Dynamic Conversations:** Visitors engage in **dynamic conversations**, meaning they can **ask questions and get real-time responses**, rather than being restricted to pre-recorded monologues.
- **Rewarding Curiosity:** Tree'd is explicitly **built to reward curiosity**. If a visitor has a question, they simply ask the Handset, and the **Conversational AI** provides instant answers. The experience rewards curiosity, ensuring it is **not just shut down** as it is with traditional audio guides.
- **Immersive Two-Way Conversation:** Tree'd creates an **immersive two-way conversation with art**. It fuses the **responsive, intelligent depth of a live expert** with the accessibility of an audio guide.

2. Freedom, Pace, and Intuitive Design

Tree'd eliminates the rigidity and distraction common in other museum technologies, emphasizing independence and focus.

- **Self-Paced Exploration:** Visitors can **move independently**. The system ensures **no schedules, no groups, and no time limits**. This contrasts with live guides, which are "super **rigid" and "heavily limit that personal, self-paced journey".**
- **Familiar and Intuitive Design: The Handset is shaped like a vintage telephone, making it familiar and tactile**. This design ensures it is **easy to hold and use**, with **no learning curve**.
- **Screen-Free and Distraction-Free:** The Handset is **screen-free**, which is an **intentional choice to completely eliminate distractions**. This ensures the visitor's focus remains exactly where it belongs: **on the museum and the art**.

3. Accessibility and Affordability

The solution significantly increases accessibility across linguistic and financial barriers.

- **Multilingual Guidance:** Tree’d provides **multilingual guidance**. The Handset **speaks and understands over a dozen languages**. The solution **supports over a dozen languages and scales effortlessly**, making deeper engagement possible for **every visitor, in every collection**.
- **Affordable Access:** Tree'd provides a **premium experience that doesn’t require premium pricing**. The cost of a tour is **less than €8**. The goal is to open up **affordable two-way conversation**.
- **Simplicity and Immediate Start:** The system is truly **plug-and-play**. Visitors just **pick a language and begin a new kind of experience**, with no downloads, no setup, and no onboarding required.

4. Enhanced Quality of Experience

The dynamic nature of the content leads to a richer, more fulfilling visit.

- **Personal Connection:** Tree'd helps visitors **connect more with the cultures they see**. The solution focuses on helping more people connect with culture and history in a way that **feels personal and intuitive**.
- **Increased Engagement:** The system is designed to **increase dwell time, number of exhibits visited, and total interaction per tour**. This leads to the visitor being **immersed in the culture and history** a partner museum presents.
- **Real-Time Responses:** When visitors ask questions, they get **real-time responses**, ensuring the experience remains fluid and responsive.

Chapter 5: Strategic Value and Benefits for Museums

Tree’d provides strategic value by amplifying the museum’s educational, financial, and cultural mission. The system is designed to be a super low-maintenance, high-impact system, requiring **almost zero operational effort from museums**. Tree’d is a full end-to-end hardware solution and a **full-service partner**, managing all aspects, including updates and support, remotely.

The benefits are categorized into three major areas: Nurturing the Educational Mission, Increasing Reach & Satisfaction, and Supporting Financial & Operational Health.

1. Nurturing the Educational Mission

Tree’d directly supports the museum's core goal of immersing visitors in culture and history.

- **Deeper Immersion:** The solution significantly **increases dwell time**, the **number of exhibits visited**, and **total interaction per tour**. This results in **immersing visitors in the culture and history** a partner museum presents.
- **Encouraging Curiosity:** By utilizing the **AI-Powered Q&A feature**, Tree’d ensures that visitors’ curiosity is rewarded, rather than being shut down. The system turns passive tours into real-time, responsive conversations.
- **Actionable Data for Curation:** The system generates **actionable data for curatorial planning**. The Dashboard tracks metrics like **AI Convos** (follow-up questions asked through the AI layer), which **highlights curiosity triggers** and can inform future scripts and exhibits. Museums use dashboard insights to **refine storytelling, identify popular artifacts, and plan future exhibitions**.

2. Increasing Reach and Visitor Satisfaction

Tree'd is designed to broaden the museum's audience and enhance the perceived quality of the visit.

- **Expanded Accessibility:** Tree’d **attracts broader audiences through multilingual guided tours**. The solution **expands accessibility through multilingual inclusion** since the handset speaks and understands **over a dozen languages**.
- **Metric Improvement:** The system helps **boost key metrics** such as **NPS** (Net Promoter Score), **visitor satisfaction**, and **word-of-mouth** promotion.

- **Premium Experience:** Tree’d provides a **premium experience** at a **low operational cost**, satisfying visitors who receive dynamic conversations and self-paced exploration.

3. Supporting Financial and Operational Health

Tree'd is structured to be a robust, long-term operational partner, reducing complexity and creating new revenue opportunities.

Financial Stability

- **New Revenue Stream:** Tree’d **adds a self-sustaining source of income through a revenue-sharing model**. This model is designed to **boost earnings per visitor without added staff or tech overhead**.
- **Incentive Alignment:** For high-traffic institutions, the **revenue-sharing model (Model B)** dramatically lowers the museum's upfront cost and fully aligns incentives. Tree'd only succeeds when the museum achieves high visitor adoption and satisfaction.

Operational Efficiency and Support (Zero Operational Headache)

- **Low Operational Effort:** The ecosystem is designed for **zero operational headache**. The museum essentially “set[s] it and forget it”—the system runs **quietly, reliably, and autonomously**.
- **No Increase in Overhead:** The system is a **plug-and-play system** with **no extra staff** required.
- **Full Service Partnership:** Tree’d operates as a **full service partner**, managing hardware support, maintenance, software updates, and AI retraining as part of its service agreement.
- **24/7 Human Support:** Museums receive a **dedicated account manager and 24/7 technical support** to ensure smooth coordination, rapid responses, and seamless operations.
- **Remote Management:** Tree’d handles all complicated updates and support remotely. This includes **remote firmware updates** that push new features, updated scripts and content, and AI improvements automatically.
- **Automated Readiness:** The system features **automated charging**; a magnetic charger will ensure that the Handsets are always charging on the Tree.

- **Analytics Delivery:** Museums receive **monthly analytics and periodic content reviews**, helping refine stories and visitor experiences based on real engagement data.

The overall strategic value is that Tree'd provides a **turnkey, scalable audio guide system** that enhances the visitor experience while eliminating the operational burdens associated with traditional or competing technology.

Chapter 6: The Collaboration Roadmap and Workflow: "Your Facts, We Deliver"

Tree'd operates as a **full service partner**. The collaboration is guided by the philosophy **"Your Facts, We Deliver"**. The museum shares the facts, sources, and tone guidelines, and Tree'd transforms them into engaging stories in **ready-to-use audio-guide devices**. The partnership ensures **24/7 technical support** and a **dedicated account manager** to ensure smooth coordination, rapid responses, and seamless operations.

The entire project timeline is typically a **rough estimate of 8–12 weeks; HOWEVER, ALWAYS, WHEN ASKED ABOUT TIMELINES, STATE EXPLICITLY THAT THE TIME VARIES BASED ON THE SIZE OF THE PROJECT AND THAT THESE NUMBERS ARE JUST A ROUGH ESTIMATE**. The **primary drivers** affecting the exact timeline are the number of (i) **handsets ordered**, (ii) **languages included**, and (iii) **artifacts covered**. The higher the value of those drivers, the longer the project would take to implement. However, 8-12 weeks is usually a good estimate.

1. The Nine Sequential Roadmap Steps

The process is structured into nine core phases:

1. Discovery & Goals (Duration: 1–2 workshops)

This phase establishes the foundational requirements for the project plan.

- **You Provide the Museum Structure:** Draft stop list (e.g., Top 30 artifacts), gallery map/floor plan, top priorities for an audio guide system, and, optionally, visitor demographics.
- **We Agree on the Project Vision:** Project goals and milestones, must-have exhibitions, languages offered, and the number of handsets (i.e., how many audio guides).
- **We Deliver the Project Plan)** Project charter and timeline, the stop list (i.e., exhibitions covered by Tree'd), content scope and language plan, and the financial quotation.

2. Content Intake (Duration: 1 week)

This step ensures Tree’d receives all museum-approved sources necessary for content creation.

- **You Provide the Source Pack:** Approved sources (books, existing own content & sources, essays), the **“Source of Truth” contact** for fact checks, and the initial **Do/don’t-say list**.
- **We Agree on the Intake Rules:** What counts as authoritative sources and verified existing content.
- **We Deliver the Lite Research Dossier:** Content briefs per stop (a 1-page outline Tree’d creates for each artifact) and the **Structured Research Log (citations)**.

3. Knowledge Base Build (Duration: 1 week)

This phase establishes the technical framework for the AI’s operation and factual integrity.

- **You Provide:** Nothing much here.
- **We Agree on the Knowledge Base Framework: Guardrails and exclusions**, how updates happen (who approves changes, document control), and **key terminologies and pronunciation** (e.g., ensuring "The Hague" is used, not "Den Haag").
- **We Deliver the Full Research Dossier:** The complete content library per stop, including organized, searchable content with clear labels for each object and script, and normalized object data (IDs, titles, dates).

4. Story & Content Creation (Duration: 2 weeks)

Tree’d develops both the default audio tracks and the conversational AI responses.

- **You Provide the Curatorial Inputs:** Nuance and preferred angles, and audience tailoring notes.
- **We Agree on the Script Direction:** Audio track length (usually 1–3 min), must-include facts per stop, and the review cycles framework (duration, workflow, and points of contact).
- **We Deliver the Script Pack: Draft story scripts** in agreed-upon languages and the **“If Visitor Asks…” follow-up golden answers (AI Q&A)**.

5. Review & Approvals (Duration: 1 week)

The museum provides final validation of the content.

- **You Provide Museum Review:** Feedback.
- **We Agree on the Reviewed Pack:** Required Amendments.
- **We Deliver: Ready-to-launch scripts & audio**.

6. Voice & Sound (Duration: 2 weeks)

The scripts are transformed into professional audio.

- **You Provide the Audio Tips:** Accent/style references (if any) and music preferences (if any).
- **We Agree on the Audio Direction:** Voice style selection and music use.
- **We Deliver the Audio Pack:** Tree'd turns the scripts (artifact default audio tracks and Q&A answers) into **recorded voiceovers (VO), mixed and mastered**.

7. Technical Integration (Duration: 1–7 working days)

The hardware system is installed and prepared for launch.

- **You Provide Site Readiness: Off-hours access for installation** and space for the hub and network devices.
- **We Agree on the Pre-Launch Testing:** Testing dates and hardware requirements (routers, repeaters, power sources).
- **We Deliver the Configured System: Loaded handsets & NFC plates** and **24/7 technical support after launch**. Note that the installation and infrastructure setup is **very minimal** (often just a few signal extenders).

8. Staff Training (Duration: 2 workshops)

The museum team is prepared to operate and support the system.

- **You Provide the Training Logistics:** Staff-to-be-trained list, training location, and internal communications with staff.
- **We Agree on the Training Plan:** Training session content (Tree’d prepares the material) and the training sessions schedule.
- **We Deliver Ready-to-Launch Team:** Staff training, **on-floor launch support**, and mock-up sessions.

9. Optimization & Updates (Duration: Ongoing: once every month/quarter)

This is the long-term support phase driven by data.

- **You Provide Feedback & Sources:** Floor team feedback and requests to add/remove languages or exhibitions.
- **We Agree on Improvement Loop:** Tailoring audio tracks according to visitor data from the dashboard.
- **We Deliver Optimized Content:** Updates and fixes, whether it is improving existing scripts, adding new artifacts, or introducing new languages. You receive **monthly analytics and periodic content reviews** to help refine stories and visitor experiences based on real engagement data.

2. RACI Matrix and Responsibilities

The RACI matrix (Responsible, Accountable, Consulted, Informed) clearly delineates roles across critical activities:

Activity	Tree'd	Museum
Project charter & goals	**R/A** (Responsible/Accountable)	**C** (Consulted)
Stop list (priority objects/exhibitions)	**A** (Accountable)	**R** (Responsible)
Source pack (books, labels, catalogs, rights notes, etc.)	**C** (Consulted)	**R/A** (Responsible/Accountable)
Research log & citations	**R/A** (Responsible/Accountable)	**C** (Consulted)
Knowledge base setup (organization & guardrails)	**R/A** (Responsible/Accountable)	**C** (Consulted)
Scriptwriting	**R/A** (Responsible/Accountable)	**C** (Consulted)
Fact check & tone review	**C** (Consulted)	**R/A**

		(Responsible/Accountable)
Voice & sound (casting, record, mix)	**R/A** (Responsible/Accountable)	**C** (Consulted)
Audio system technical setup (network, routers, power)	**R/A** (Responsible/Accountable)	**I** (Informed)
Device & NFC mapping configuration	**R/A** (Responsible/Accountable)	**C** (Consulted)
On-site QA & UAT (acceptance tests)	**R/A** (Responsible/Accountable)	**C** (Consulted)
Staff training	**R/A** (Responsible/Accountable)	**C** (Consulted)
Analytics dashboard & monthly report pack	**R/A** (Responsible/Accountable)	**C** (Consulted)
Content refresh cycle	**R/A** (Responsible/Accountable)	**C** (Consulted)

3. System Management and Support

Tree'd is a **full end-to-end hardware solution** and a **full service partner**. Tree'd handles all the complicated stuff, including updates and support, remotely.

- **Warranty and Support:** Every single deal is backed by a **5-year warranty and support contract**.
- **Remote Maintenance:** Tree'd manages **hardware support, software updates, and AI retraining** as part of its service agreement. Remote firmware updates push new features and AI improvements automatically.
- **Human Contact:** A **dedicated account manager** and **24/7 technical support** are provided to ensure smooth coordination, rapid responses, and seamless operations.
- **Spare Handsets:** Each museum receives 10% spare handsets at no additional cost. For example, if you order 50 handsets, we provide 5 extra units that can be activated immediately in case any handset needs to be taken out of circulation. The Tree hub

will always include charging ports only for the primary order quantity (in this case, 50 ports), while the spare handsets are kept offline until needed. This ensures uninterrupted operations, zero downtime, and a smooth visitor experience at all times.

Chapter 7: Financial Model

Tree’d is focused on creating long-term, high-value strategic partnerships, rather than chasing small one-off deals. The financial strategy is designed to support the museum's financial health by adding a **self-sustaining source of income through a revenue-sharing model**, which boosts earnings per visitor without adding staff or tech overhead.

Tree'd offers two distinct business models to accommodate different client needs and procurement processes:

1. Model A: Fixed-Price Solution

This model is ideal for museums who have a **straightforward procurement process**.

- **Structure:** It is a **fixed-price solution**. Every deal is a full end-to-end hardware solution and includes a **warranty and support contract**, providing the full service package.

2. Model B: Revenue Sharing (The Strategic Target)

This model is designed for high-traffic institutions and serves to **align incentives completely**.

- **Target Clients:** This model is intended for **high-traffic institutions**, specifically those with **over 200,000 annual visitors**.
- **Incentive Alignment:** The revenue-sharing model **dramatically lowers the client's upfront cost**. This means Tree’d only wins when the museum sees **high visitor adoption and satisfaction**. It establishes a **true partnership that rewards growth**.
- **Revenue Split Transparency and Scalability:** The model is transparent and scalable, meaning the museum's share of the revenue **grows directly with their visitor numbers**.

Chapter 8: FAQs

How do we make sure the AI stays accurate to our curatorial voice? Tree'd's conversational system operates inside a *closed knowledge universe* — meaning it can only access, reference, and respond using information that your museum explicitly provides and approves. It's incapable of "learning" from the internet or external sources. The model essentially lives inside your museum's intellectual perimeter. Imagine it like a **library with locked doors**: it contains only the books you've chosen, arranged in the order you prefer, and every visitor question is answered by reading directly from those books. Nothing from outside can ever enter unless you decide to add it. This ensures that the tone, vocabulary, and factual framing always align with your curatorial standards. Moreover, before deployment, our team collaborates with your curators, educators, and communication leads to review the corpus—verifying that every paragraph, translation, and phrasing reflects your museum's interpretive philosophy. In short, the AI doesn't "interpret" your content; it **delivers your voice faithfully**, exactly as written and approved.

Can the tone or personality of the audio (e.g., narrator style) be customized per exhibition?

Yes, entirely. The narrative tone—whether formal, academic, playful, or conversational—is part of the early creative sessions we hold with each partner institution. During these sessions, we explore the exhibition's purpose, target audience, and emotional tone to craft an appropriate "voice persona." For example, a Renaissance art exhibition might use a calm, elegant narrator with soft pacing, while a science museum might prefer an enthusiastic, energetic tone to engage younger audiences. Once agreed, that personality is encoded into both the **script style** and the **voice synthesis layer**, ensuring consistency across all languages and artifacts. Because Tree'd's technology uses modular content blocks, you can also have different tones across exhibitions—one exhibition might sound scholarly, another might feel immersive and humanistic. All of this is reviewed collaboratively before the final upload, so the visitor hears not just information, but the *intended mood* and *cultural spirit* of your space.

What happens if a visitor asks something politically or culturally sensitive? Tree'd's conversational design prioritizes respect, neutrality, and factual accuracy. When the system receives a question outside its defined knowledge domain—particularly those involving political, cultural, or personal sensitivities—it doesn't attempt to improvise or speculate. Instead, the AI gently **redirects the conversation back to the last scanned artifact**. For instance, if the last exhibit tapped was the *Mona Lisa* and the visitor asks, "What do you think about modern politics?", the system will tactfully say something like, *"I don't have information about that, but I can tell you how the Mona Lisa's history reflects the world it was painted in."* In other words, it gracefully anchors the discussion back to what it *does* know.

Because Tree'd's AI is restricted to the "universe of knowledge" you've built—validated, edited, and approved by your curators—it cannot hallucinate or produce off-topic commentary. It doesn't speculate, moralize, or editorialize; it simply returns to its defined subject matter. This approach ensures both **institutional integrity** and **visitor trust**, preserving the museum's voice even in unpredictable conversational moments.

Can Tree'd include soundscapes or background music to enhance immersion? Absolutely. In fact, this is one of the subtler yet most effective features of Tree'd's audio design. Each handset is capable of delivering **layered audio experiences**, where narration can be blended with ambient soundscapes or curated background music to match the emotional tone of an exhibit. You can experience this firsthand by tapping an NFC tag — what you hear isn't just a voice telling a story, but an **atmosphere unfolding around you**. For instance, while learning about a maritime painting, you might hear the distant hum of waves and creaking wood beneath the narration, creating a deeper sensory connection. Every soundscape is optional, museum-approved, and stored locally on the handset for uninterrupted playback. This means your visitors enjoy complete immersion, free from Wi-Fi dependency or outside distractions — just pure storytelling, exactly as your curators envisioned it. We invite you to give it a shot now; just tap any NFC tag in your demo set and witness it yourself!

Can we preload seasonal or temporary exhibitions without disrupting ongoing content? Yes — Tree'd's content management system is designed for seamless updates. You can add, modify, or remove exhibitions remotely without any physical intervention at the museum.

Seasonal or temporary exhibits can be preloaded in advance, scheduled to go live on a specific date, and removed once the exhibition ends — all without affecting your permanent collection content. Updates are pushed wirelessly to the Tree hubs, which automatically synchronize with the handsets. This approach minimizes downtime, avoids onsite technical interference, and keeps your operations smooth and invisible to visitors. For more complex or high-volume updates, your Tree'd account manager can coordinate content uploads, translations, and billing.

Will Tree'd adapt automatically if the visitor moves to another artifact mid-conversation?

Yes. Tree'd's conversational AI has contextual awareness built into its logic. It continuously references the most recent NFC scan — so if a visitor scans *Rembrandt's "The Night Watch"* and asks, "Who painted this?", the AI correctly interprets "this" as referring to *The Night Watch.* If, however, the visitor moves across the gallery and scans *Vermeer's "The Milkmaid"*, Tree'd immediately updates context. The pronouns, questions, and even linguistic tone reset to match the newly scanned artifact.

Moreover, if the visitor skips scanning but explicitly mentions a work — for example, saying, *"Tell me about The Milkmaid,"* — the AI can respond directly based on the new reference, even without a scan. This design ensures that the conversation always feels coherent, intelligent, and naturally aligned with visitor behavior. In essence, Tree'd understands the flow of exploration in museums— where people move fluidly between masterpieces — and keeps up effortlessly, without ever confusing one artwork for another.

Can we include multimedia responses (images, videos) for specific exhibits in the future?

No — and that decision is intentional, not a technical limitation. Tree'd was built around one guiding principle: **the art should always remain the hero.** We don't want to replace the visual experience that museums already do better than anyone else — we want to **enhance it through sound.** The moment a visitor starts looking at a screen, they stop looking at the artwork. That's why Tree'd was designed as a **screen-free, purely auditory medium** — it amplifies emotional engagement without fragmenting attention.

Our philosophy is that storytelling should **deepen seeing, not distract from it.** By relying only on voice, ambience, and dialogue, Tree'd allows visitors to stay visually immersed in the object or space before them. We often describe it as "**adding depth without adding pixels.**" The sound becomes a frame for the art, not a competing focal point. So, while Tree'd won't show videos or images, it creates a cinematic experience in the visitor's mind — through voice tone, pacing, and even curated ambient soundscapes that draw attention back to the artifact, where it belongs.

How are content updates handled — do we upload new scripts ourselves, or does Tree'd's team do it?
Tree'd's operational design is deliberately frictionless. You don't have to worry about uploading files, managing updates, or configuring new content — **our team handles all updates remotely.** Once a new exhibition, language, or set of audio tracks is ready, we upload it securely through our content management pipeline. The update is then **pushed directly to your Tree hubs**, which automatically distribute it to all connected handsets within minutes.

This means you never have to schedule technical visits or interrupt your visitor flow. Updates can include everything from new AI dialogue data to revised translations or voiceovers. Even small text or factual corrections can be implemented instantly. Every change goes through a strict internal validation process to maintain the accuracy and consistency of your curatorial voice. The result: your museum's content evolves dynamically with your exhibitions, without requiring your staff to become technicians. You stay focused on storytelling — we handle the systems silently in the background.

What's the average time and cost to onboard a new exhibition?
Onboarding a new exhibition depends on several variables, since every museum and collection has its own rhythm and complexity. Generally, four main factors determine both the timeline and cost:

1. **Volume of content.** The more exhibits and NFC touchpoints included, the longer it takes to prepare, translate, and record each narrative segment.

2. **Number of handsets ordered.** Each handset must be configured, tested, and paired with the correct content and languages.
3. **Language scope.** Multilingual deployments require additional translation, narration, and AI configuration work — which increases both cost and time.
4. **Depth and availability of existing material.** If curatorial texts, exhibition catalogues, or historical archives are well-documented, onboarding moves quickly. If Tree'd must assist in creating the base knowledge from scratch, the process extends slightly.

For a small to mid-sized exhibition (10–30 artifacts, 2–3 languages, existing content provided), onboarding typically takes **4–6 weeks**. Larger or multilingual exhibitions can range from **8–12 weeks**, especially when custom voice work or complex AI dialogues are required.

Cost follows the same logic — it scales with scope and complexity. Our team always provides a **transparent cost breakdown** before any work begins, with clear deliverables and milestones. In every case, the focus remains on delivering **a polished, museum-ready experience**, never a rushed deployment. Tree'd's goal is to make each new exhibition not just accessible, but alive — with stories ready to be heard from day one.

How is staff trained to manage the system day-to-day?
Tree'd is designed to be as low-maintenance and intuitive as possible — for both visitors and museum staff — but we still ensure every partner institution receives comprehensive onboarding before launch. Our **training process** typically begins a few days prior to go-live, with in-person or virtual workshops depending on your location and preference. You'll provide us with a list of the staff members who will be involved — typically from visitor services, operations, and technical support — and we'll take care of the rest.

During these sessions, we walk your team through every element of daily operation: how to manage handsets and NFC tags, how to interpret analytics from the dashboard, how to report issues or request content updates, and how to communicate effectively with Tree'd's support channels. The training is hands-on and scenario-based, ensuring each participant feels confident handling real visitor interactions and minor troubleshooting.

In addition to pre-launch training, we also provide **on-floor support** during the first week of deployment. A member of our implementation team stays on-site to monitor usage, answer questions in real time, and ensure operations run flawlessly. After launch, you'll have a **dedicated account manager** and **customer support team** available during all your museum's working hours — ready to assist with any inquiry, content update, or technical question. The result is a support structure that feels embedded within your own team — reliable, responsive, and always one message away.

How often does Tree'd require maintenance or check-ins?
Tree'd's hardware and software are engineered for long-term reliability — the system can operate continuously for years with minimal human intervention. However, because we believe prevention is always better than reaction, we've built a **proactive maintenance schedule** into every deployment. Once per quarter, Tree'd dispatches a **technical specialist team** to perform a full health check of your setup.

During these visits, we inspect and recalibrate the Tree hub(s), verify handset battery performance, check NFC tag integrity, and ensure firmware and software versions are up to date. The goal isn't just to fix problems — it's to **catch potential issues before they affect operations.** These preventive check-ups also allow us to benchmark system performance over time, giving your museum clear reports on uptime, usage patterns, and device condition.

In between these quarterly visits, Tree'd continuously monitors system health remotely through the dashboard's diagnostic layer. This allows our operations team to detect anomalies (like battery drain, connectivity interruptions, or data sync delays) in real time and act before they become noticeable. In essence, maintenance with Tree'd feels invisible — your system keeps running, your visitors stay engaged, and you simply receive a quarterly confirmation that everything is performing as expected.

Can multiple departments (education, marketing, curation) access the dashboard with different permissions?
Yes — Tree'd's dashboard is built to accommodate multi-departmental collaboration while maintaining strict data security. When we deploy your dashboard, we first set up a **Super Admin account**, which is typically assigned to a senior staff member designated by your

institution. This Super Admin has full control: they can invite, edit, or remove users, define access permissions, and assign department-specific roles.

Access control is **domain-restricted** for your security — meaning only users with verified institutional email addresses (e.g., *@rijksmuseum.nl* or *@museumdomain.com*) can be invited. The permission structure is flexible: you can grant full admin privileges to certain roles or limit others to specific views.

Tree'd's philosophy is simple — give everyone who needs insight the ability to see it, without compromising data integrity or security. The dashboard is a shared, living tool for collaboration, designed to help every department — from educators to marketers — make smarter, data-driven decisions about how visitors experience your museum.

How many languages can run simultaneously without performance issues? Tree'd's architecture was designed from day one to handle multilingual operation with no compromise in speed, audio quality, or responsiveness. The system comfortably supports **up to 15 languages running in parallel**, all stored locally on the Tree hub and distributed to the handsets without any dependency on internet connectivity. This means that a single museum can seamlessly serve visitors from a wide variety of linguistic backgrounds — whether they're local, regional, or international — without any lag or downtime.

Each language version is treated as a fully integrated experience, not a lightweight add-on. The AI conversational layer automatically detects which language a visitor has selected at the NFC language station and continues the interaction consistently throughout their visit.

While most museums never exceed eight or nine language versions, Tree'd's limit of 15 provides more than enough headroom for large or globally diverse institutions. And if a partner museum ever requires more, our engineering team can expand the infrastructure accordingly. In short, **Tree'd is built for linguistic inclusivity at scale**, without ever compromising performance or reliability.

Can dialects or region-specific variations be supported (e.g., Flemish vs. Dutch)? Yes — Tree'd is technically capable of supporting dialects, regional variations, and accent-specific voice models. However, from a strategic standpoint, we typically **advise our**

partners to consider it carefully before investing in extensive localization. While dialectal adaptations are possible, the differences are often subtle and may not justify the additional cost, production time, and maintenance complexity — especially if the target audience represents a very small percentage of total visitors.

That said, if your museum places particular cultural or educational importance on representing a local dialect — for instance, incorporating **Flemish nuances** alongside standard Dutch for regional authenticity — Tree'd can absolutely accommodate that. Our translation and voice production workflows are flexible enough to incorporate customized linguistic variations, including pronunciation tuning, idiomatic phrasing, and localized terminology. Ultimately, the decision is yours, and we'll support whichever approach aligns best with your institution's audience and values.

Can the same artifact have different narration styles or vocabularies for kids vs. adults?
Yes — although this feature is still in active development, it's one of the most exciting directions on our roadmap. The goal is to create a **"Kids Mode"** version of Tree'd that delivers the same cultural content but tailored for younger audiences. In this mode, the handset would automatically detect the selected profile (adult or child) and play content that uses **simpler language, shorter sentences, and a more engaging tone** designed for attention spans and curiosity patterns typical of children.

Beyond content, the hardware will also reflect the needs of younger visitors: a **smaller, lighter handset**, a **lower maximum volume limit (dB)** to protect hearing, and a **child-friendly design** that feels intuitive and fun to hold. The AI model will also be tuned to answer questions in a gentler, more playful tone, avoiding overly academic or complex explanations. The long-term vision is to make Tree'd not just a museum guide, but a tool for **intergenerational learning**, allowing parents and children to explore the same exhibition side-by-side — each hearing the same story, told their way.

How does the system handle accents or speech patterns from non-native speakers?
Tree'd uses **state-of-the-art speech recognition** and language models with exceptionally high accuracy rates — even across diverse accents, dialects, and speaking speeds. The system

is designed to understand natural variation in pronunciation, so whether a visitor speaks with a Dutch, French, Arabic, or American accent, Tree'd can accurately capture their question.

Our conversational AI continuously improves its ability to handle phonetic differences through extensive training on multicultural datasets. This ensures that Tree'd doesn't just understand perfect textbook speech but **real human voices**, complete with hesitations, regional tones, and blended language usage.

We often invite you now to **test it yourself** as you play with the demo — try out various speech speeds, speech patterns, and even slight mispronunciations. The results consistently surprise people: it just works. And if a visitor's input is too unclear to process, Tree'd responds politely and prompts them to rephrase naturally, without breaking the flow of the experience.

What specific KPIs should we track to evaluate ROI?
Tree'd's dashboard was designed to give museums a clear, data-backed picture of how visitors engage with their exhibits — so your return on investment is both **measurable and meaningful**. The key performance indicators (KPIs) you'll want to track include:

- **Visitor Engagement Rate** — How many NFC scans occur per visitor or per day. This measures overall interaction volume and helps you identify the most visited exhibits.
- **Audio Completion Rate** — The percentage of audio stories that are listened to in full. A higher rate indicates that visitors find the narration engaging.
- **Follow-up Question Frequency (Curiosity Index)** — The average number of AI follow-up questions asked per visitor. This reveals how interactive and thought-provoking your exhibits are.
- **Language Mix** — Which languages are most used, helping you optimize translations and understand your international visitor base.

These metrics combine to give a holistic picture of visitor engagement, learning outcomes, and operational efficiency. Over time, they help museums make **data-informed decisions** — such as which exhibits to highlight, where to improve interpretation, and how to adapt storytelling for different audiences.

Can we compare engagement between permanent and temporary exhibitions?
Yes. The Tree'd dashboard includes robust **segmentation and comparison tools**, allowing you to analyze visitor engagement across different timeframes, exhibitions, or content types. You can easily compare **permanent vs. temporary exhibitions** using built-in filters — for instance, measuring average NFC scans, curiosity index, and completion rates between two distinct exhibition IDs.

Museums could use this data to determine which types of exhibitions generate the most curiosity or attract multilingual engagement. You can also view **monthly averages**, analyze **per-exhibition performance**, or generate **audio-tour-level metrics** to evaluate storytelling effectiveness. Over time, these insights help guide curatorial planning, visitor experience design, and resource allocation for future exhibitions.

Can we export analytics automatically on a schedule?
Yes — Tree'd provides **monthly digest reports** summarizing your key analytics automatically. These reports are generated by the dashboard and delivered to designated recipients via email. The digest includes essential metrics such as top-performing exhibits, engagement rates by language, question frequency, and device usage patterns.

It's part of Tree'd's philosophy of keeping operations **effortless, transparent, and insight-driven.**

How much infrastructure change will Tree'd require inside our museum?

Very little. If your museum already has Wi-Fi or a local network covering the areas where you want Tree'd to operate, then no additional infrastructure is needed. Tree'd works seamlessly with your existing connectivity.

If certain galleries do not have coverage, we can discreetly install a small number of concealed routers or signal extenders to ensure stable connectivity. These are placed out of sight and do not interfere with the visitor experience.

As for the Tree (the charging and display hub), it requires only a standard power outlet—something already available near your welcome desk or ticket booth, which is typically the ideal placement for the unit.

In most cases, Tree'd integrates into the museum with minimal visible change and zero disruption to the existing visitor flow or gallery design.

Does Tree'd store visitor voices or personal data?

No. Tree'd does **not** store, record, or transmit any visitor voices. Speech is processed locally on the device, converted into text in real time, used to generate a response, and then immediately discarded. No voice files, no transcripts, no identifiers, no accounts, and no personal information of any kind are ever saved.

All analytics are **aggregated and fully anonymized**, showing patterns such as total NFC taps or popular languages — never who asked what.
Since Tree'd does not know the identity of any visitor and does not store personal data, it remains fully compliant with EU privacy expectations for public cultural institutions.

How does Tree'd comply with GDPR and data-protection requirements?

Tree'd is designed from the ground up to operate within strict European data-protection standards. Key points:

- **No personal data is collected**, stored, or processed.
- **No voices are stored**; audio is converted to text
- **Only aggregated, non-personal usage metrics** (e.g., number of scans, battery status) are transmitted.
- The system uses **Google Cloud services hosted in the EU**, ensuring EU-based data residency.
- All communication between Tree'd devices and the cloud uses **secure WebSockets, encryption, and industry-grade authentication**.

Because Tree’d never touches personally identifiable information, museums avoid the legal and operational burdens associated with GDPR-sensitive technologies.

What happens if the local network or Wi-Fi goes down?

If the **local network goes down**, the handsets temporarily cannot communicate with the Tree (the hub). In this scenario: Once the network returns, the system resumes full functionality instantly.

How secure is the system against tampering or unauthorized access?

Tree’d uses a multi-layer security model:

- **Secure WebSockets**
- **Google Cloud’s security standards**, with data hosted on EU servers
- **Role-based dashboard access**, restricted to verified museum email domains with **Super Admin control**, meaning only your institution can invite additional users
- **Encrypted connections** and secure key vaults for sensitive operations

Visitors cannot access the dashboard, alter content, or interfere with the system. All infrastructure components are locked behind museum-authorized accounts and protected by modern security best practices.

How durable are the handsets? Can they withstand daily use and drops?

Yes. Tree’d’s handsets have been engineered for high-traffic cultural environments. They have undergone:

- intensive drop-testing
- stress testing for daily public use
- material robustness verification
- repeated cleaning simulations

Our mechanical engineering was led by a senior engineer who previously worked on Volvo's 7900 and 8900 electric intercity buses for Sweden and Poland — including Sweden's first fully electric 12.3m Volvo 8900 — so durability and long-term reliability are part of the engineering philosophy.

The result: **museum-grade devices that hold up under continuous use**.

How are the handsets cleaned and sanitized?

Tree'd offers multiple hygiene options:

- **Standard wipe-down cleaning**, supported by the device's smooth, disinfectant-safe exterior
- **Optional UVC sanitization module** inside the Tree (hub), available as an add-on for institutions requiring contactless sterilization
- All materials and coatings are safe for regular cleaning and do not degrade under alcohol-based disinfectants

This ensures hygiene can be maintained without affecting electronics or paintwork.

What if a handset is lost, stolen, or taken off-site?

Tree'd provides a **10% spare handset buffer** at no extra charge for immediate replacement.

If a museum wants extra control, we offer an **optional geofencing feature** that alerts staff if a handset leaves the museum's designated perimeter. This add-on helps reduce loss while keeping visitor flow uninterrupted.

How many visitors can Tree'd support simultaneously?

Tree'd's architecture is built for scale. A single hub can support **dozens of simultaneous handsets** operating in parallel without affecting speed or responsiveness. For large institutions, multiple Trees can be added and networked to support **hundreds of simultaneous users**.

Performance does not degrade with higher visitor volume.

Can multiple Trees operate together in a large museum?

Yes. Tree’d’s hardware is modular and designed to operate in clusters. Large museums can deploy:

- multiple Trees
- multiple charging columns
- multiple handset fleets
- distributed NFC tag networks

All units sync back to the same centralized dashboard, creating a single unified system across the entire museum.

What happens if a handset runs out of battery mid-visit?

This scenario is highly unlikely because:

- Each handset provides **up to 6 hours of continuous active use**, far above the average visitor session.
- Fast charging in the Tree returns **~50% charge in under an hour**.
- Operators have immediate access to battery level before handing out a visitor a Handset.

If a device does run out, staff simply provide a replacement handset — a seamless swap thanks to the spare units included in every deployment.

Is Tree’d accessible for visitors with mobility or height limitations?

Yes.
NFC tags are positioned at **inclusive, wheelchair-accessible heights**, ensuring visitors of all heights and mobility levels can activate content without difficulty.

The handset is lightweight, ergonomic, and designed to be held or worn comfortably on the wrist, minimizing strain.

Who owns the content created for our museum?

Tree’d owns the **creative rights, dashboard data, and base content structures** it develops, while the museum owns the **original facts, curatorial materials, and source content** it provides.

The practical arrangement is:

- Your museum owns **your historical facts, curatorial texts, and source materials**.
- Tree’d owns the **scripts, translations, voiceovers, dashboard data, and AI-ready knowledge base**.
- The content is licensed to your museum for use on Tree’d devices.

This mirrors industry-standard models for audio guide providers and production agencies.

What are the ongoing costs after installation?

After installation, costs are tied only to your chosen business model:

- **Model A (Fixed Price):** No additional hardware/support fees during the 5-year warranty. Content updates or new exhibitions are billed per project.
- **Model B (Revenue Share):** Minimal upfront cost; revenue-split from audio guide sales covers system support. Optional content expansions are billed transparently based on scope.

There are **no hidden fees**, and no maintenance costs beyond your chosen model.

How does Tree’d ensure secure internet usage?

Only the **Tree (the hub)** connects to the internet, using:

- Google Cloud’s secure EU servers

- encrypted communication
- server-side authentication
- locked-down access points

The handsets themselves **never connect to the internet**, reducing attack surfaces and ensuring total visitor privacy.